Paper Review 6: Multimodal Neurons in Artificial NNs

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Question 1:

Neurons can activate strongly for certain inputs if that input is highly representative of certain features of the data in the data set. Mathematically, the input is multiplied with different weight vectors for specific neurons. If one of those linear combinations for the weight vector results in a higher value, that neuron will produce a large positive output, which will then be passed through an activation function and through the rest of the model. A high positive value created a 'strong activation', signaling to the rest of the network that the specific concept or feature has been detected.

Question 2:

As the weights get aligned to match the concept they are emphasizing for the neuron, the backpropagation updates that weight in order to minimize the loss of the output. Gradient decent is about using the gradient of the weight against loss to update the weight to go towards the loss minima. If the gradient is negative, the loss minima is further down the line, so the weight should be updated in a positive value. If the gradient is positive, the loss minima was passed, so the weight should be updated backwards. This is process achieved through backpropagation, and the gradients of parameters closer to the beginning of the model depend on the gradients of the parameters closer to the end of the model. This updating of parameters with gradients ensures that the weigh values stay in a rage where they are actually describing concepts in a meaningful way that helps create a more accurate output.

Two sentence summary: The model does not begin by wanting to identify any specific concept, but it learns over time that failing to identify that concept leads to higher loss. Backpropagation then molds the weight matrices to help create a neuron that can detect the concept.

Question 3:

Throughout training, the weight vector get aligned to match the concept/feature that neuron is creating the 'strong activation' for. This vector alignment happens through the combination of the forward and back propagation processes. Where high positive outputs are allowed to pass through the model, affecting the end result and the loss, and negative outputs are stopped by the activation function, not being allowed to affect the end result and loss. Then the loss gets calculated, and the gradients are passed back through the model to change the weight so that they create outputs that are more correct. This process of strong activation and updating weights leads to weight vectors that align with input vectors to highlight useful features. There is no manual feature engineering in this process. This is an advantage because sometimes humans do not know what features are available to choose, so the manual process is not available. For example, the authors notices that there were Africa neurons that can differences beyond the features they were aware of, but they did not have the cultural background to know what they were. It is also easier to use a model that can learn how to code for a specific feature, especially if that feature is very complex and can't be shown graphically.

Question 4:

The model needs to be optimized to minimize loss. If the input takes multiple types of formats, the model can be inclined to learn how to include both formats to make more accurate outputs. Neural networks learn representations in hidden layers using vector math. The neurons that exist earlier in a model handle all the "physical" attributes of the input (e.g. edges in pixels, or recognizing letters in text). As more hidden layers are passed, the neurons begin to mathematically explain the meaning that is represented in the input, regardless of input format. Therefore, inputs that are of the same concept, but different formats get mapped to the same region in vector space, and the neuron for that concept is activated when either format is used as input.

Statement of AI:

I used Google Gemini to help me with this assignment. I read the papers on my own and tried to answer them on my own first. Then I gave Gemini the question and my answer and asked it to tell me if I missed any key topics or misunderstood the paper. I also asked it to tell me where in the text I should re-read. I used this response to help me edit my answer and help my understanding of the papers.